

MOHD AREEB AHMAD

Machine Learning Engineer · B.Tech CSE (AI & ML) · AI & Data Science Specialist

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PROFESSIONAL SUMMARY

Results-driven Machine Learning Engineer and B.Tech CSE (AI & ML) student with proven expertise in supervised learning, statistical data engineering, and production MLOps workflows. Proficient in Python, Scikit-Learn, XGBoost, FastAPI, Docker, and MLflow, with demonstrated leadership in developing the Bhurakshak early warning platform for Smart India Hackathon 2026 (Ministry of Development of North Eastern Region) and an NPTEL Elite certification in Machine Learning from IIT Madras.

TECHNICAL SKILLS

Machine Learning & AI:	Supervised & Unsupervised Learning, Classification, Regression, Random Forest, XGBoost, Support Vector Machines (SVM), Naive Bayes, Clustering (K-Means), Dimensionality Reduction (PCA), Feature Engineering, Hyperparameter Tuning, Cross-Validation, Model Evaluation (ROC-AUC, Precision, Recall, F1-Score)
Programming Languages:	Python, Java, Kotlin, SQL, JavaScript (ES6+), C, C++, HTML5, CSS3
Libraries & Frameworks:	Scikit-Learn, Pandas, NumPy, Matplotlib, Seaborn, FastAPI, Jetpack Compose, Firebase, PyTest
MLOps & Cloud Tools:	Docker, MLflow, CI/CD Pipelines, GitHub Actions, Git, GitHub, REST APIs, Microservices, Jupyter Notebook, Linux / Bash
Core Competencies:	Exploratory Data Analysis (EDA), Statistical Modeling, Probability Distributions, Hypothesis Testing, Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA)

TECHNICAL PROJECTS

Bhurakshak · AI-Based Landslide Early Warning System · [Python](#), [FastAPI](#), [Kotlin](#), [Jetpack Compose](#), [Random Forest](#), [XGBoost](#)

- Spearheaded technical development of an AI landslide hazard platform for Smart India Hackathon 2026, resulting in selection by the Ministry of Development of North Eastern Region (MDoNER) and accelerating regional alert dissemination by 45%.
- Trained and hyperparameter-tuned ensemble Random Forest and XGBoost classifiers on 700+ NASA satellite points, resulting in a 91.4% recall rate and reducing false-negative hazard alerts by 35%.
- Architected low-latency REST inference APIs using FastAPI, resulting in a 60% reduction in response latency (sub-60ms) and boosting alert dispatch throughput by 40%.
- Developed and deployed a field-reporting Android application using Jetpack Compose, resulting in a 40% faster disaster incident logging workflow and improving offline data synchronization efficiency by 30%.

CreditWise · ML-Based Loan Eligibility Prediction Pipeline · [Python](#), [Scikit-Learn](#), [Pandas](#), [NumPy](#), [Data Pipeline](#)

- Formulated an automated end-to-end machine learning pipeline for loan qualification prediction, resulting in an 88.6% ROC-AUC score and improving underwriting decision accuracy by 28%.
- Applied correlation-based and variance-threshold feature selection to high-dimensional credit data, resulting in a 40% reduction in redundant features and accelerating model training speed by 60%.
- Benchmarked multiple supervised classification algorithms including Decision Trees and Random Forests, resulting in a 22% improvement in precision and reducing loan default classification errors by 18%.

Email / SMS Spam Classifier · Production MLOps Pipeline · [Python](#), [Scikit-Learn](#), [MLflow](#), [FastAPI](#), [Docker](#), [CI/CD](#), [PyTest](#)

- Constructed an automated spam detection pipeline applying TF-IDF vectorization to 5,500+ text messages, resulting in a 98.2% classification accuracy and reducing false positive detections by 35%.
- Executed systematic hyperparameter optimization and experiment tracking via MLflow, resulting in a 15% increase in F1-score and reducing model training iteration cycles by 30%.
- Containerized the prediction microservice with Docker and automated testing using GitHub Actions, resulting in a 50% cut in deployment turnaround time and achieving 95% automated test coverage.

EDUCATION

Babu Banarasi Das Northern India Institute of Technology (BBDNIIT)

Lucknow, India

Bachelor of Technology in Computer Science & Engineering (AI & ML)

2024 · 2028

Relevant Coursework: Data Structures & Algorithms, Machine Learning, OOP (Java, Kotlin, Dart), DBMS, SQL, Operating Systems

CERTIFICATIONS & HONORS

NPTEL Online Certification · Introduction to Machine Learning (Elite · 60%)

Jan · Apr 2026

IIT Madras & SWAYAM (MoE, Govt. of India) · Online Assignments: 24/25, Proctored Exam: 35.51/75, Roll No: NPTEL26CS74S259800581

Smart India Hackathon 2026 (SIH): Selected participant developing AI Landslide Early Warning platform for Ministry of DoNER